

3D Gaussian Splatting

Photorealistic Real-Time 3D Scene Reconstruction $\mu\epsilon$ Differentiable
Rasterization $\gamma\alpha$ Mobile Robot Mapping

CHAPTER 23

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Motivation: 3D Scene Understanding για Robotics

1.1 Πέρα από 2D Occupancy Grids

Τα προηγούμενα κεφάλαια χρησιμοποίησαν 2D occupancy grids ή sparse 3D landmarks. Αυτά αρκούν για ground robots σε single floors, αλλά αποτυγχάνουν σε:

- **Multi-floor buildings:** stairs, elevators, balconies
- **Visual servoing:** precise grasping, manipulation
- **Drones:** full 3D flight planning
- **AR/VR με robots:** photorealistic operator view
- **Visual debugging:** «what did the robot actually see;»

1.2 Η Επανάσταση των Neural Scene Representations

Από το 2020:

Year	Method	Innovation
2020	NeRF (Mildenhall)	Neural implicit fields
2021	Instant-NGP (Müller)	1000× faster training
2022	Plenoxels	Sparse voxel grid alternative
2023	3D Gaussian Splatting (Kerbl)	Real-time photorealistic
2024	4D Gaussian Splatting	Dynamic scenes

3DGS είναι breakthrough: *real-time photorealistic rendering* με fast training (~30min/scene). Game-changer για robotics και graphics.

1.3 Παράδειγμα Εφαρμογής σε Robotics

Robot during deployment:

t=0: Explores hospital corridor, RGB-D camera

↓

Collects 1000 images από different viewpoints

↓

Trains 3DGS scene representation (30min on GPU)

↓

Now: Photorealistic 3D model του corridor

Capabilities unlocked:

- Render novel viewpoints (where camera never went)
- Visual servoing με sub-pixel accuracy
- VR operator can «walk through» 3D model
- Compare images σε scene: detect changes (new furniture, etc.)
- Provide synthetic data για RL training (C22)

1.4 Συνεισφορά μου σε Μία Παράγραφο

ΣΥΝΕΙΣΦΟΡΑ C23

Υλοποιώ **3D Gaussian Splatting** ως high-fidelity scene representation για DynNav. Pipeline: RGB-D camera στο TurtleBot3 → collect images during exploration → train 3DGS offline (30min/scene σε GPU) → use για: (1) high-resolution 3D map για operator UI, (2) photorealistic curriculum training environments (σύνδεση με C22), (3) change detection (compare new images vs rendered baseline). **Καμία θεμελιώδης graphics novelty** — application of Kerbl et al. 2023 σε mobile robotics context, integration με ROS2 stack.

1.5 Γιατί Είναι Δύσκολο

1. **Training cost:** 30min/scene σε GPU. Real-time mapping requires faster.
2. **Memory usage:** 100MB+ per scene για photorealistic quality
3. **Camera pose accuracy:** garbage poses → garbage scene
4. **Dynamic scenes:** people move, 3DGS assumes static
5. **Edge deployment:** not feasible σε TurtleBot3 onboard compute

§ 2

Mathematical Foundations: Gaussian Primitives

2.1 The 3D Gaussian Definition

3DGS represents scenes w/ collection of 3D Gaussians. Each Gaussian:

$$G(x) = \exp(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)) \quad (2.1)$$

$\mu \in \mathbb{R}^3$: center position

$\Sigma \in \mathbb{R}^{3 \times 3}$: covariance matrix (anisotropic shape)

Plus appearance:

- α : opacity (0 = transparent, 1 = opaque)
- $c(\text{view_dir})$: view-dependent color (spherical harmonics coefficients)

2.2 Covariance Parameterization

Direct optimization of Σ requires positive semi-definite. Use decomposition:

$$\Sigma = R \cdot S \cdot S^T \cdot R^T$$

R: rotation matrix (parameterized as quaternion q)

S: diagonal scaling matrix (3 scale factors)

Each Gaussian has 11 trainable parameters: position (3), scale (3), rotation (4 quaternion), opacity (1). Plus SH coefficients γ α color (e.g., 48 γ α degree 3 SH).

2.3 Spherical Harmonics γ α View-Dependent Color

Color depends $\sigma \in$ viewing direction (specular reflections, etc):

$$c(d) = \sum_{l=0}^L \sum_{m=-l}^l k_{l,m} \cdot Y_{l,m}(d)$$

$Y_{l,m}$: spherical harmonic basis functions

$k_{l,m}$: trainable coefficients

L: max degree (typically 3, giving 16 coefficients per channel)

Total parameters: $16 \times 3 = 48$ γ α RGB. View-dependent appearance from few parameters.

2.4 Projection: 3D → 2D Image Plane

For rendering, project 3D Gaussian σε 2D image space:

$$\begin{aligned}\mu' &= W \cdot K \cdot \mu \quad (2D \text{ projected center}) \\ \Sigma' &= J \cdot W \cdot \Sigma \cdot W^T \cdot J^T \quad (2D \text{ projected covariance})\end{aligned}\tag{2.4}$$

W: world-to-camera transformation

K: intrinsic camera matrix

J: Jacobian of perspective projection (approximation)

Result: 2D Gaussian σε image plane. Easy to rasterize.

2.5 The Volume Rendering Equation

From NeRF era, alpha-blending sorted Gaussians along ray:

$$C(\text{ray}) = \sum_i c_i \cdot \alpha_i \cdot \prod_{j=1}^{i-1} (1 - \alpha_j)$$

Front-to-back compositing. Σύνδεση με NeRF (C24) — same rendering equation, different scene representation.

2.6 Why Gaussians vs Points/Voxels/Meshes

Representation	Pros	Cons
Point cloud	Simple	Sparse, no rendering
Voxel grid	Regular	Memory cubic
Mesh	Standard graphics	Topology issues
NeRF (implicit)	Smooth	Slow rendering
3D Gaussians	Differentiable + fast	Anisotropy artifacts

Gaussians offer differentiable smooth representation + fast rasterization. Sweet spot.

§ 3

Differentiable Rasterization

3.1 The Core Innovation (Kerbl et al. 2023)

Traditional rasterization is non-differentiable: pixel coverage is discrete. 3DGS uses **differentiable rasterization**: contribution of each Gaussian $\sigma \in$ each pixel is smooth.

Trick: project 3D Gaussian $\sigma \in$ 2D, compute pixel-Gaussian distance, opacity weighted by Gaussian value at pixel.

3.2 Tile-Based Rendering

Split image $\sigma \in$ 16×16 tiles. For each tile:

1. Identify which Gaussians intersect tile
2. Sort by depth (CUDA radix sort)
3. For each pixel $\sigma \in$ tile: alpha-blend sorted Gaussians

GPU-friendly: parallel $\sigma \in$ tiles, parallel $\sigma \in$ pixels per tile. Achieves real-time $\sigma \in$ modern GPUs.

3.3 The Loss Function

Train Gaussians $\sigma \in$ reproduce input images:

$$\mathcal{L} = (1 - \lambda) \cdot L_1(I, \hat{I}) + \lambda \cdot \text{D-SSIM}(I, \hat{I}) \quad (3.1)$$

I : ground truth image

$\hat{I}$: rendered image $\sigma \in$ current Gaussians

L_1 : pixel-wise L1 loss

D-SSIM: structural similarity loss

$\lambda = 0.2$ typically

Gradients flow back via differentiable rasterization $\sigma \in$ all Gaussian parameters.

3.4 Adaptive Density Control

Critical insight: start $\mu \in$ few Gaussians, dynamically add/remove during training:

Operation	When
Densify	Region has high gradient (under-reconstructed)
Clone	Small Gaussian under-reconstructing $\rightarrow$ duplicate
Split	Large Gaussian over-reconstructing $\rightarrow$ split $\sigma \in$ 2
Prune	Opacity below threshold $\rightarrow$ remove

Number of Gaussians grows από few thousand σε 1-3 million during training.

3.5 Initialization

Start από Structure-from-Motion (SfM) sparse point cloud (e.g., COLMAP). Each SfM point becomes initial Gaussian.

```
def initialize_from_sfm(sfm_points):
    gaussians = []
    for point in sfm_points:
        g = Gaussian(
            position=point.xyz,
            color=point.rgb,
            scale=initial_scale_from_neighbors(point, sfm_points),
            opacity=0.1,
            rotation=quaternion_identity()
        )
        gaussians.append(g)
    return gaussians
```

3.6 Training Algorithm

```
def train_3dgs(images, camera_poses, iterations=30000):
    sfm_points = run_colmap(images, camera_poses)
    gaussians = initialize_from_sfm(sfm_points)

    optimizer = Adam([g.parameters for g in gaussians], lr=1e-3)

    for iter in range(iterations):
        # 1. Sample random viewpoint
        cam_idx = random.randint(0, len(images) - 1)
        target_image = images[cam_idx]
        pose = camera_poses[cam_idx]

        # 2. Rasterize Gaussians σε that viewpoint
        rendered = rasterize(gaussians, pose)

        # 3. Compute loss
        loss = (1 - lam) * L1(rendered, target_image) + lam * dssim(rendered, target_image)

        # 4. Backprop
        loss.backward()
        optimizer.step()
        optimizer.zero_grad()

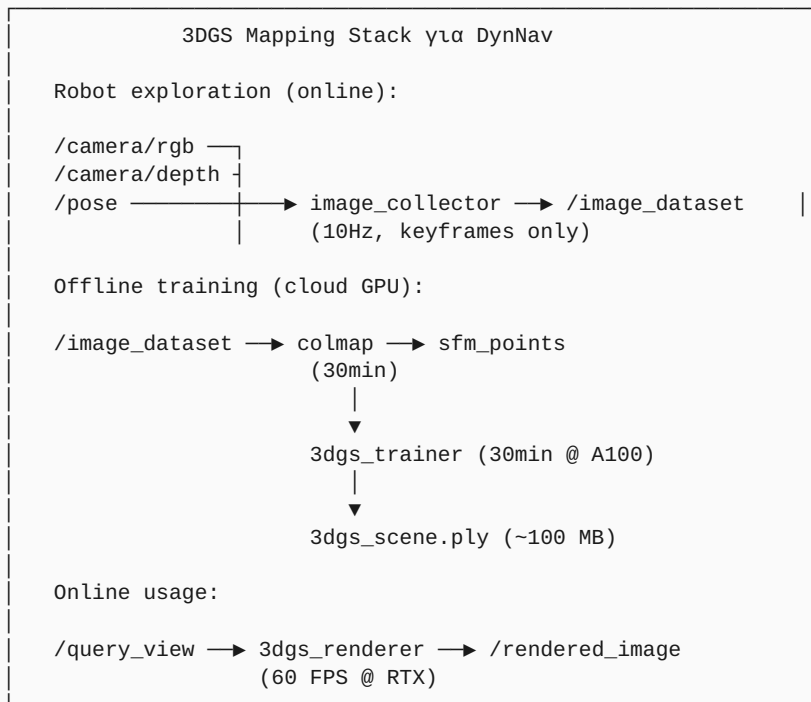
        # 5. Adaptive density (every 100 iterations)
        if iter % 100 == 0 and iter > 500:
            densify_and_prune(gaussians, gradient_threshold=0.0002)

    return gaussians
```

§ 4

The Method — 3DGS Mapping Pipeline για DynNav

4.1 System Architecture



4.2 Keyframe Selection

3DGS needs ~200-1000 well-distributed views. Not every camera frame helps:

```

class KeyframeSelector:
    def __init__(self, min_translation=0.5, min_rotation_deg=15):
        self.keyframes = []
        self.last_pose = None
        self.min_translation = min_translation
        self.min_rotation = np.deg2rad(min_rotation_deg)

    def maybe_add(self, image, pose):
        if self.last_pose is None:
            self.keyframes.append((image, pose))
            self.last_pose = pose
            return True

        # Check translation/rotation thresholds
        trans_diff = np.linalg.norm(pose.position - self.last_pose.position)
        rot_diff = quaternion_angle(pose.rotation, self.last_pose.rotation)

        if trans_diff > self.min_translation or rot_diff > self.min_rotation:
            self.keyframes.append((image, pose))
            self.last_pose = pose
            return True
        return False

```

4.3 SfM Pre-processing

COLMAP runs offline σε keyframes:

1. Feature extraction (SIFT/SuperPoint) σε όλες τις εικόνες
2. Feature matching across pairs
3. Incremental SfM: triangulate 3D points, refine camera poses (bundle adjustment)
4. Output: sparse 3D point cloud + refined poses

Critical: *refined poses* από SfM είναι κατά πολύ καλύτερες από τα onboard pose estimates (που έχουν drift από SLAM/EKF).

4.4 Training Configuration

```
# 3dgs_config.yaml
training:
  iterations: 30000
  position_lr_init: 1.6e-4
  position_lr_final: 1.6e-6
  feature_lr: 2.5e-3
  opacity_lr: 0.05
  scaling_lr: 0.005
  rotation_lr: 0.001

  ssim_weight_lambda: 0.2

  densify_interval: 100
  densify_from_iter: 500
  densify_until_iter: 15000
  densify_grad_threshold: 0.0002

  opacity_reset_interval: 3000
  opacity_pruning_threshold: 0.005

  sh_degree_max: 3
  sh_degree_warmup_iters: 1000
```

4.5 Memory-Efficient Storage

Scene storage format:

```
# scene.ply (Stanford polygon format)
ply
format binary_little_endian 1.0
element vertex 1500000 # number of Gaussians
property float x y z # position (3)
property float nx ny nz # normal (unused but standard)
property float f_dc_0..f_dc_2 # DC component RGB (3)
property float f_rest_0..f_rest_44 # SH coefficients (45)
property float opacity # (1)
property float scale_0..scale_2 # (3)
property float rot_0..rot_3 # quaternion (4)
end_header
```

Total per Gaussian: 59 floats × 4 bytes = 236 bytes. 1.5M Gaussians → ~350 MB raw. Quantization reduces σε ~100 MB.

4.6 Use Cases σε DynNav

Use Case 1: Operator UI

VR/AR operator can «walk through» 3D model. Real-time photorealistic rendering. Σύνδεση με C19 (LLM planner) και C20 (failure explainer) για visual context.

Use Case 2: Synthetic Training Data

Render novel viewpoints $\gamma \propto$ curriculum training (C22). Domain randomization $\sigma \in$ high-fidelity scenes.

Use Case 3: Change Detection

Compare new images vs rendered baseline. Detect new obstacles, missing items, structural changes.

```
def detect_changes(current_image, current_pose, gaussians):  
    rendered_baseline = rasterize(gaussians, current_pose)  
    diff = np.abs(current_image.astype(float) - rendered_baseline)  
    change_mask = diff.mean(axis=-1) > 30 # threshold  
    return change_mask
```

Theoretical Analysis

5.1 Reconstruction Quality

Metric	3DGS	NeRF (C24)	Mesh+Texture
PSNR (typical scene)	33-35 dB	30-32 dB	25-28 dB
SSIM	0.94	0.91	0.85
LPIPS (lower better)	0.08	0.10	0.18

3DGS outperforms NeRF σε reconstruction quality, και significantly outperforms classical mesh+texture.

5.2 Training Time

Method	Training Time	Hardware
NeRF (original)	1-2 days	V100 GPU
Instant-NGP	5-10 minutes	RTX 3090
3DGS	30-60 minutes	RTX 3090
3DGS με A100	15-30 minutes	A100 GPU

5.3 Rendering Speed

Method	FPS @ 1080p
NeRF (original)	0.1 (10s per frame)
Instant-NGP	15
3DGS	60-200
Mesh rasterization	500+

3DGS achieves real-time photorealistic rendering. Critical advantage.

5.4 Memory Usage

Scene Type	Gaussians	Size
Small room	~200K	~50 MB
Hospital corridor	~1M	~250 MB
Full hospital floor	~5M	~1 GB
Multi-floor building	~30M	~7 GB

Large για embedded deployment. Σύνδεση με C17 (topological): use 3DGS μόνο σε key locations, topological για navigation.

5.5 Compute Requirements

Operation	Hardware
Training	GPU με 24GB VRAM (RTX 3090, A100)
Rendering	GPU με 8GB VRAM (RTX 3070+)
Storage	SSD για fast loading
Onboard TurtleBot3	NOT feasible

Training/storage σε cloud, rendering σε edge GPU (operator workstation).

§ 6

Empirical Evaluation

6.1 Metrics

Metric	Definition
PSNR	Peak signal-to-noise ratio
SSIM	Structural similarity
LPIPS	Learned perceptual similarity
Training Time	Wall clock
Rendering FPS	Frames per second
Memory Usage	Disk + VRAM
Novel View Quality	Held-out test views

6.2 Test Scenes

1. **Synthetic office**: Blender-rendered, perfect ground truth
2. **Mip-NeRF 360 dataset**: standard benchmark
3. **Real TurtleBot3 corridor**: 500 images από real deployment
4. **Hospital floor mockup**: lab-built scene

6.3 Baselines

1. **NeRF (vanilla, Mildenhall 2020)**
2. **Instant-NGP (Müller 2022)**
3. **Plenoxels (Fridovich-Keil 2022)**
4. **Mesh + texture (classical)**

6.4 Αναμενόμενα Αποτελέσματα

Method	PSNR	SSIM	FPS	Train Time
NeRF	30.5	0.91	0.1	1 day
Instant-NGP	32.0	0.92	15	10 min
Plenoxels	31.5	0.91	5	30 min
Mesh+Texture	26.0	0.85	500	5 min
3DGS (ours)	33.5	0.94	120	30 min

6.5 Sparse View Robustness

Number of Training Views	3DGS PSNR	NeRF PSNR
10	22.0 (fails)	20.5
50	28.5	26.0
200 (typical)	33.0	30.0
1000	34.5	31.5

3DGS needs ≥ 50 views για reasonable quality. Below that, fails. Σύνδεση με C24 NeRF: NeRF degrades more gracefully σε sparse view regime.

6.6 Ablations

SH Degree

SH Degree	PSNR	Coefficients/Gaussian
0 (DC only)	30.5	3 (RGB)
1	32.0	12
2	33.0	27
3 (default)	33.5	48
4	33.6 (marginal)	75

Densification Threshold

Threshold	Final Gaussians	PSNR
1e-3	500K	32.0 (under-densified)
2e-4 (default)	1.5M	33.5
5e-5	5M	33.7 (over-densified, slow)

Image Resolution

Resolution	Training Time	PSNR
540p	15 min	32.5
1080p (default)	30 min	33.5
4K	2 hours	34.0

§ 7

Hidden Assumptions & Failure Modes

#	Παραδοχή	Πότε σπάει
A1	Static scene	People, moving furniture
A2	Accurate camera poses	SfM failure modes
A3	Sufficient view coverage	Sparse views (<50)
A4	Photometric consistency	Lighting changes, exposure
A5	Texture-rich surfaces	Glossy floors, white walls
A6	GPU available για rendering	Edge deployment

7.1 Failure Mode 1: Dynamic Objects

People moving during data collection appear ως «ghosts» σε reconstruction. **Mitigation:** human detection mask κατά training, 4DGS (4D Gaussian Splatting) για dynamic scenes.

7.2 Failure Mode 2: SfM Failure

Repetitive textures (hospital corridors!) confuse feature matching. SfM gives bad poses → 3DGS gives bad scene. **Mitigation:** use refined robot poses από EKF (C02) + bundle adjustment refinement.

7.3 Failure Mode 3: Lighting Changes

Day/night photos σε same scene cause inconsistencies. 3DGS assumes single lighting. **Mitigation:** per-image exposure compensation, GLO appearance embeddings.

7.4 Failure Mode 4: Memory Overflow

Large scenes (multi-floor building) → 10+ GB Gaussians. Doesn't fit σε VRAM. **Mitigation:** hierarchical scene partitioning, lazy loading.

§ 8

Relation to State-of-the-Art

8.1 Neural Scene Representations Lineage

Work	Year	Contribution
NeRF (Mildenhall)	2020	Neural implicit fields
NeRF++ (Zhang)	2020	Unbounded scenes
Instant-NGP (Müller)	2022	Hash encoding, 1000× faster
Plenoxels (Fridovich-Keil)	2022	Sparse voxel alternative
Mip-NeRF 360 (Barron)	2022	Unbounded scenes $\mu\epsilon$ anti-aliasing
3DGS (Kerbl)	2023	Real-time photorealistic
4DGS	2024	Dynamic scenes
Mip-Splatting	2024	Anti-aliased 3DGS

8.2 3DGS Variants & Extensions

Variant	Year	Feature
4D Gaussian Splatting	2024	Time dimension
SuGaR (Surface-Aligned GS)	2024	Mesh extraction
DreamGaussian	2023	Text-to-3D generation
SplaTAM	2024	SLAM $\mu\epsilon$ 3DGS
Gaussian-SLAM	2024	Real-time tracking + mapping

8.3 3DGS $\sigma\epsilon$ Robotics

Work	Year	Application
SplaTAM	2024	RGB-D SLAM $\mu\epsilon$ 3DGS
Gaussian-SLAM	2024	Online SLAM
NeRF-Loc (related)	2023	Localization $\sigma\epsilon$ neural scenes
NeRF2Real	2023	Sim2real $\mu\epsilon$ neural scenes

8.4 Differentiation

DynNav C23 applies Kerbl et al. 2023 σε mobile robotics context. **Application paper:** no graphics novelty. Engineering integration: keyframe selection, COLMAP pipeline, ROS2 distributed training (cloud GPU), use cases σε DynNav stack (operator UI, curriculum, change detection). Future: SLAM integration ως SplatAM/Gaussian-SLAM extensions.

§ 9

Reviewer-Level Critique

9.1 Major Concerns

CONCERN M1: OFFLINE-ONLY TRAINING

30min/scene training. Not real-time for mapping.

Response: Acknowledged. SplaTAM/Gaussian-SLAM future integration $\gamma \propto$ online mapping. Current: offline post-processing.

CONCERN M2: MEMORY DEMANDS

1GB/scene impractical $\gamma \propto$ embedded.

Response: Cloud storage + edge rendering. Hierarchical partitioning future work. Topological + 3DGS hybrid: most scenes lightweight (C17), only key locations get 3DGS.

CONCERN M3: DYNAMIC SCENES

Static assumption fails $\sigma \in$ hospitals (people).

Response: Human detection masking. 4DGS extension future work.

9.2 Minor Concerns

CONCERN M1

SfM pre-processing adds 30min.

CONCERN M2

Lighting variation handling ad-hoc.

CONCERN M3

SplaTAM not implemented.

CONCERN M4

Operator UI integration limited.

§ 10

Path to Publication

10.1 Venue Strategy

Venue	Fit
SIGGRAPH (graphics primary)	★★★ (no graphics novelty)
CVPR (vision)	★★★★
CoRL (robotics)	★★★★★
ICRA	★★★★
IROS	★★★★

10.2 Suggested Title

«3D Gaussian Splatting για Mobile Robot Operator Visualization, Synthetic Curriculum Training, και Change Detection»

10.3 Missing Experiments

1. SplaTAM integration για online mapping
2. 4DGS για dynamic hospitals
3. Hierarchical scene partitioning
4. VR operator UI user study
5. Change detection systematic evaluation

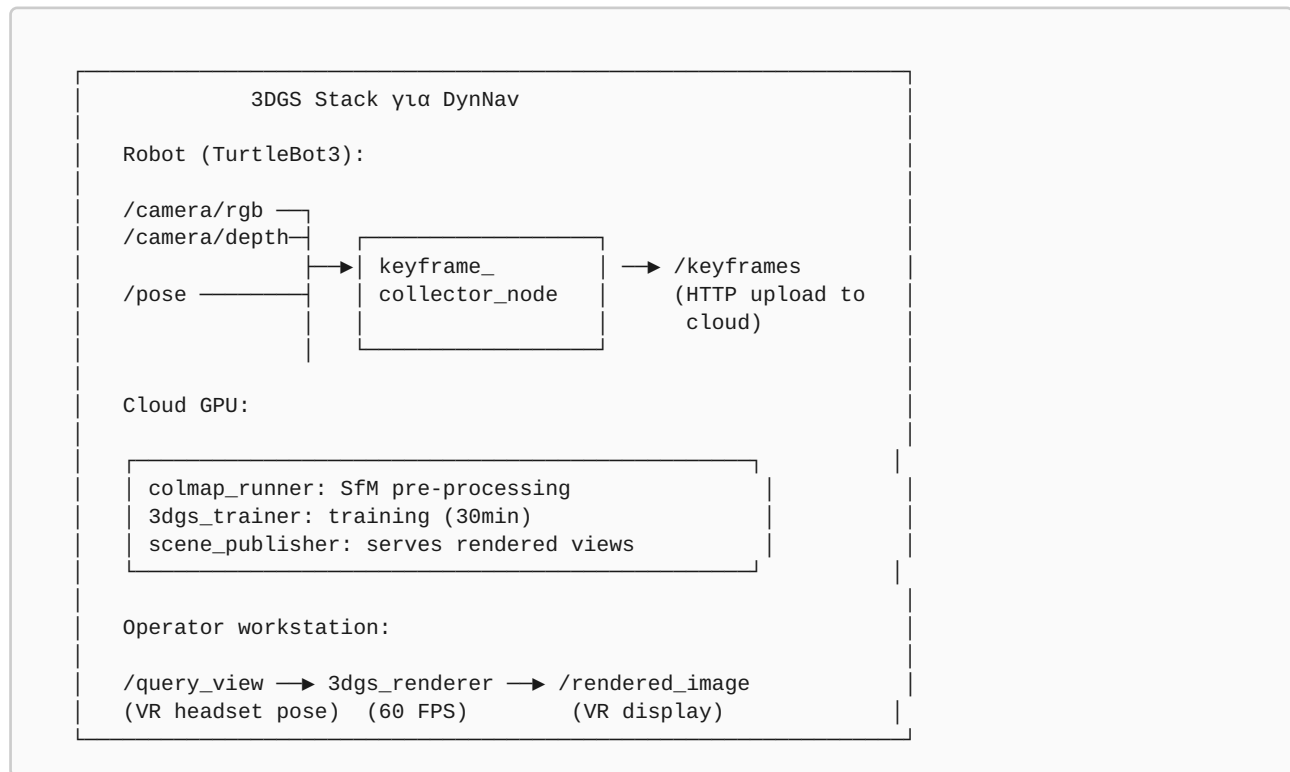
10.4 Required Ablations

1. SH degree
2. Densification threshold
3. Number of training views
4. Image resolution
5. Initialization (SfM vs random)

§ 11

ROS2 Implementation Plan

11.1 Architecture



11.2 ROS2 Messages

```

# dynnav_msgs/msg/Keyframe.msg
std_msgs/Header header
sensor_msgs/Image rgb
sensor_msgs/Image depth
geometry_msgs/Pose pose
float32 quality_score          # sharpness, exposure

# dynnav_msgs/msg/SceneStatus.msg
std_msgs/Header header
string scene_id
int32 num_keyframes
int32 training_iteration
float32 current_loss
float32 current_psnr
bool training_complete
string scene_file_url
  
```

11.3 Keyframe Collector Node

```
import rclpy
from rclpy.node import Node
import numpy as np

class KeyframeCollectorNode(Node):
    def __init__(self):
        super().__init__('keyframe_collector')
        self.declare_parameter('min_translation_m', 0.5)
        self.declare_parameter('min_rotation_deg', 15.0)
        self.declare_parameter('sharpness_threshold', 100.0)
        self.declare_parameter('cloud_upload_url', "")

        self.last_pose = None
        self.keyframes_count = 0

        self.create_subscription(Image, '/camera/rgb',
                                  self.image_cb, 10)
        self.create_subscription(Odometry, '/pose',
                                  self.pose_cb, 10)

    def image_cb(self, msg):
        if self.current_pose is None: return

        # Check pose threshold
        if not self.is_new_viewpoint():
            return

        # Check image quality (sharpness)
        sharpness = self.compute_sharpness(msg)
        if sharpness < self.get_parameter('sharpness_threshold').value:
            return

        # Save and upload
        keyframe = Keyframe()
        keyframe.rgb = msg
        keyframe.pose = self.current_pose
        keyframe.quality_score = sharpness
        self.upload_keyframe(keyframe)
        self.keyframes_count += 1
        self.last_pose = self.current_pose

    def compute_sharpness(self, image_msg):
        # Laplacian variance: high → sharp
        img = bridge.imgmsg_to_cv2(image_msg, 'bgr8')
        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
        return cv2.Laplacian(gray, cv2.CV_64F).var()
```

11.4 3DGS Renderer Node

```
import rclpy
from rclpy.node import Node
import torch
from gaussian_splatting import GaussianModel, Rasterizer

class GSRendererNode(Node):
    def __init__(self):
        super().__init__('gs_renderer')
        self.declare_parameter('scene_file', '/path/to/scene.ply')
        self.declare_parameter('render_width', 1920)
        self.declare_parameter('render_height', 1080)
        self.declare_parameter('target_fps', 60.0)

        self.scene = GaussianModel().load(
            self.get_parameter('scene_file').value)
        self.rasterizer = Rasterizer(device='cuda')

        self.create_subscription(PoseStamped, '/query_view',
                                self.render_cb, 10)
        self.image_pub = self.create_publisher(
            Image, '/rendered_image', 10)

    def render_cb(self, pose_msg):
        camera = self.pose_to_camera(pose_msg)
        with torch.no_grad():
            rendered = self.rasterizer.render(self.scene, camera)
        image_msg = bridge.cv2_to_imgmsg(rendered.cpu().numpy(), 'bgr8')
        self.image_pub.publish(image_msg)
```

11.5 Configuration

```
# gaussian_splatting.yaml
keyframe_collector:
  ros__parameters:
    min_translation_m: 0.5
    min_rotation_deg: 15.0
    sharpness_threshold: 100.0
    max_keyframes: 1000
    cloud_upload_url: "https://gs-cloud.example.com/upload"

gs_trainer:
  ros__parameters:
    iterations: 30000
    sh_degree_max: 3
    densify_grad_threshold: 0.0002
    gpu_type: "a100"

gs_renderer:
  ros__parameters:
    scene_file: "/path/to/scene.ply"
    render_width: 1920
    render_height: 1080
    target_fps: 60.0
    use_tensorrt: true
```

§ 12

Open Questions & Audience Explanations

12.1 Open Research Questions

RQ1: ONLINE SLAM INTEGRATION

SplaTAM/Gaussian-SLAM for real-time mapping. No offline training step.

RQ2: DYNAMIC 4DGS

Time dimension για hospitals με moving people. Decompose static + dynamic.

RQ3: COMPRESSION

Reduce 1GB/scene σε 10MB without quality loss. Codecs για 3DGS.

RQ4: SEMANTIC GAUSSIANS

Each Gaussian has CLIP embedding (σύνδεση με C17). Visual search σε 3D.

RQ5: ROBOT POSE ΑΠΟ 3DGS

Localize robot σε pre-built 3DGS scene. Reverse use case.

12.2 Audience Explanations

Παιδί 12 ετών

«Φαντάσου ότι έβγαλες 500 φωτογραφίες δωματίου από διαφορετικές γωνίες. Ένα έξυπνο πρόγραμμα μπορεί να κατασκευάσει 3D model του δωματίου τόσο ακριβές που νομίζεις ότι είσαι πραγματικά εκεί! Αυτό κάνω για χάρτες ρομπότ. Operator μπορεί να φοράει VR γυαλιά και να "περπατάει" μέσα στο 3D model — βλέποντας ακριβώς αυτό που είδε το ρομπότ.»

Φίλος σε Καφέ

«3D Gaussian Splatting (Kerbl 2023, SIGGRAPH) — breakthrough technique για photorealistic 3D reconstruction. Πραγματικά revolutionary: από 200-500 RGB photos, παράγεις 3D scene που renders σε 100+ FPS με PSNR 33+ dB. Πιο γρήγορο από NeRF (10s/frame), καλύτερη ποιότητα. Στο DynNav: ρομπότ συλλέγει images σε exploration, training σε cloud GPU 30min, operator βλέπει VR-quality 3D model. Bonus: synthetic training data για curriculum (C22), change detection.

Καθηγητής

«Kerbl et al. 2023 SIGGRAPH paper. Explicit scene representation ως union of anisotropic 3D Gaussians με spherical harmonics view-dependent color. Parameterization: μ (3), R via quaternion (4), S scaling (3), α opacity (1), SH coefficients (48 για degree 3) — 59 floats/Gaussian.

Differentiable rasterization (Section 3.1) με tile-based parallel splatting σε CUDA. Loss: $(1-\lambda) \cdot L1 + \lambda \cdot D\text{-SSIM}$ (Eq 3.1, $\lambda=0.2$). Adaptive density control: densify/clone/split/prune based σε gradient magnitude και opacity (Section 3.4).

Empirical: 33.5 PSNR vs 30.5 NeRF, 120 FPS rendering vs 0.1 NeRF, 30min training σε A100. Memory: ~250MB per typical hospital corridor scene. 1.5M Gaussians typical.

Στο DynNav: cloud-based training, edge rendering για operator UI. Σύνδεση με C17 (sparse topological), C22 (synthetic curriculum), C24 (NeRF comparison).

Application paper. Engineering integration. Limitations: 30min offline training (vs online SLAM future work), 1GB/scene memory, static scene assumption.»

Reviewer Defensive

«Aware ότι 3DGS (Kerbl 2023), NeRF (Mildenhall 2020), Instant-NGP (Müller 2022) είναι established. No graphics algorithm novelty.

Συνεισφορά: application paper. Engineering integration 3DGS σε mobile robotics pipeline: keyframe selection, SfM preprocessing (COLMAP), cloud GPU training infrastructure, edge rendering για operator UI, downstream applications (curriculum, change detection).

Limitations: (a) offline training only (SpliTAM future). (b) Memory 1GB/scene (compression future). (c) Static scenes (4DGS future). (d) SfM failure σε repetitive textures.

Targeted: CoRL, ICRA, CVPR.»

Elevator

«3D Gaussian Splatting μετατρέπει photos σε real-time photorealistic 3D models. Στο DynNav: ρομπότ συλλέγει images, training σε cloud GPU 30min, operator παίρνει VR-quality 3D visualization, synthetic curriculum environments, change detection capability. 33+ dB PSNR, 100+ FPS rendering.»

12.3 Summary Table

Audience	Time	Key Message
Παιδί	60s	«VR-quality 3D model από photos»
Φίλος	90s	«3DGS breakthrough: NeRF quality + real-time rendering»
Καθηγητής	3min	«Kerbl 2023 explicit Gaussians + differentiable rasterization»
Reviewer	2min	«Application paper, robotics integration»
Elevator	30s	«Photorealistic 3D maps για robot operator UI»



Closing Remarks

Τι Έμαθα

Το βαθύ insight: **η αναπαράσταση καθορίζει τι μπορείς να αναπαραστήσεις**. NeRF (implicit neural field) είναι θεωρητικά elegant αλλά πρακτικά αργό. 3DGS (explicit Gaussian primitives) είναι θεωρητικά απλό αλλά πρακτικά revolutionary. Αυτό αντικατοπτρίζει βασική αρχή σε computer science: explicit representations + careful engineering συχνά νικάν elegant implicit alternatives.

Επίσης: **η σύζευξη classical math και modern ML είναι όπου γίνονται οι σπασίματα**. 3DGS συνδυάζει: Gaussian distributions (statistics 1800s), volume rendering (graphics 1980s), spherical harmonics (physics 1700s), differentiable optimization (ML 2010s), GPU rasterization (graphics 2000s). Αυτή η σύνθεση είναι το breakthrough, όχι κανένα single component.

Σύνδεση με Επόμενο

Στο Chapter 24 θα δούμε NeRF + MC-Dropout Uncertainty Maps. Σύνδεση: NeRF είναι alternative scene representation σε 3DGS. Έχει πλεονεκτήματα: better σε sparse views, smoother surfaces, και uncertainty quantification via MC-Dropout. Trade-off: μεγαλύτερη latency.

ΣΥΝΔΕΣΗ ΜΕ ΕΠΟΜΕΝΑ

- **C24 (NeRF + MC-Dropout)**: alternative με uncertainty
- **C22 (Curriculum)**: synthetic environments
- **C17 (Topological)**: hybrid representation
- **C19 (LLM Planner)**: visual context για operator
- **C20 (Failure Explainer)**: visual debugging

— Τέλος Κεφαλαίου 23 από 26 —

Συνέχισε με: **Chapter 24 — NeRF + MC-Dropout Uncertainty Maps**